Element E

E1: STEM Principles

Our highest priority is a moderate flow rate, so we want to first identify how much storage and force we will need for this.

Another top priority is portability. We will first identify common weights of possible portable sinks, then determine if this is realistic for a teacher or student aid. When taking into account the weight of the sink parts we believe that it would be optimal for the teacher to take the water holding tank out and replace them back into the sink to mitigate the weight of the total sink all in one, but rather separating the weight that the teacher would transport and a given time.This would also take away the bulky-ness of taking and filling up the whole storage container.

In addition to this, student functionality is extremely important to us, for obvious reasons. We will analyze the force needed to turn the handle, heights of the typical user, and other functionality aspects.

E2: STEM Principles Application

Flow Rate: To have an adequate flow rate we would need to control the diameter of the hole of which the water comes out of. To increase flow rate it is necessary to decrease the size of the diameter. In order to have an adequate flow rate we would have to take into account the diameter of the hole in order to make sure that the water flows well and up to our standard. Another thing we must take into account is if we are using an additional slope to help move the water into the sink bowl then we must take into account that the water sliding will decrease the flow rate. In order to apply this accurately then we will need to utilize mathematics in order to calculate the flow rate in advance to mitigate problems down the road.

Another important aspect of the sink is cleanliness and ease to clean. Special education classrooms can become dirty easily at a faster rate than regular classrooms. The teachers at Whitemarsh Elementary have a large array of cleaning equipment in the classroom to help mitigate the overall dirtiness of the classroom. To comply with the standards of the teacher, the sink must also have a cleanliness element to it. To ensure we meet that quota of cleanliness it is necessary to test the growth of bacteria and use materials that mitigate the bacterial growth. Bacteria growth is most stimulated when it is in a warm and moist environment, the sink itself naturally creates a wet environment which alone can stimulate the growth of harmful bacteria. Keeping the temperatures down in the sink is going to be harder as the inside of the storage container will get hot. Utilizing materials such as stainless steel will be integral in building the design as it is a good material in not wearing down to harsher chemicals which will help with the cleaning of the sink. Also bacteria will not be able to grow as easily if the chemicals such as unscented chlorine bleach to wipe out most harsh bacteria. Materials such as plastics and metal at the same time deter bacteria growth as they are inorganic materials.

Expert	Credentials
John Sciotto jr.	- Bachelor's degree in Civil Engineering from Penn State University - Owned an engineering firm - Director of Preconstruction - AVP of Preconstruction and Project Controls
Clint Rickert	- Masters degree in Technical Education and Agriculture from University of California, Pennsylvania - Bachelor’s degree in Science Education from Widener University - Science and technical education teacher for 22+ years
Mr. Muscarella	- Bachelor’s degree in Chemical Engineering from Drexel University - Master’s degree in Science of Instruction - Resident Engineering teacher, teaching for 28+ years